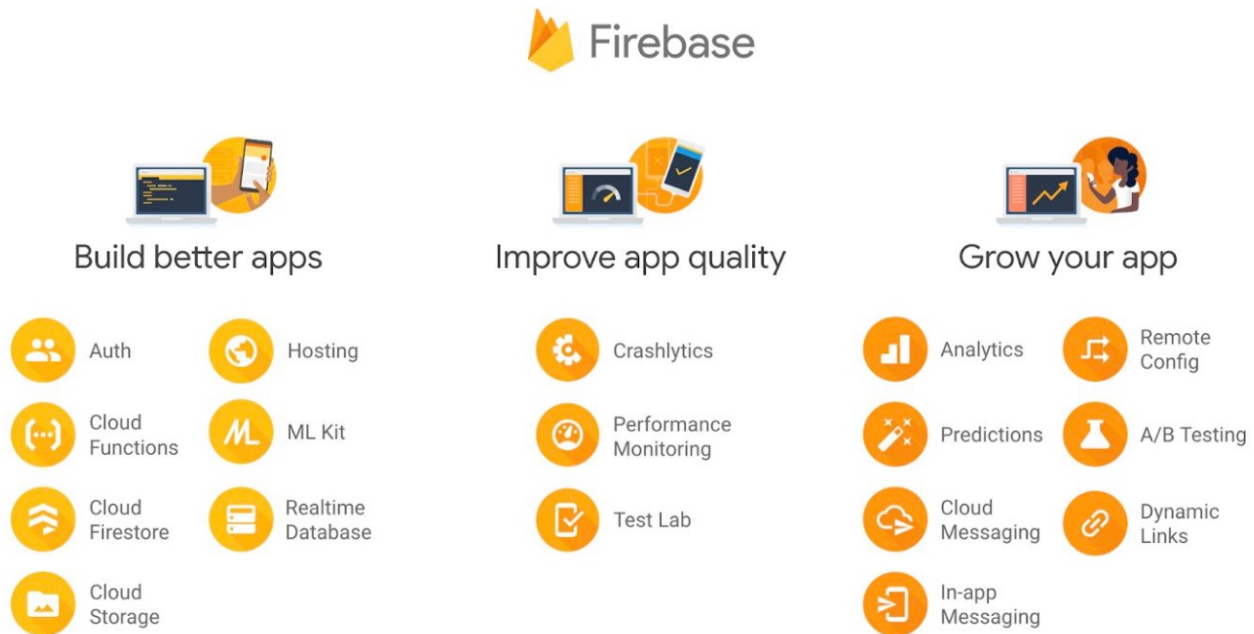


Firestore

CMPUT 301

What is Firestore?

- Firestore is a cloud-hosted NoSQL database
- It is part of Firebase, Google's app development platform
- Apps can use Firestore to save, load, and synchronize data
- Firestore is commonly used for mobile and web apps
- Firebase provides many services
 - Firestore is the database service used in this lab



Relational vs. Document Databases

- Relational databases organize data into tables, rows, and columns
- Firestore organizes data into collections, documents, and fields
- A document is similar to one record or object
- A collection is a group of documents

Relational Database	Firestore
Table	Collection
Row	Document
Column	Field

Documents

- A document stores field and values
- Each field has a name
- Values can be strings, numbers, booleans, lists, maps, and timestamps
- Each document has an ID
- A document can store up to 1MiB of data

Document ID: city_001

name: "Edmonton"

province: "AB"

population: 1010000

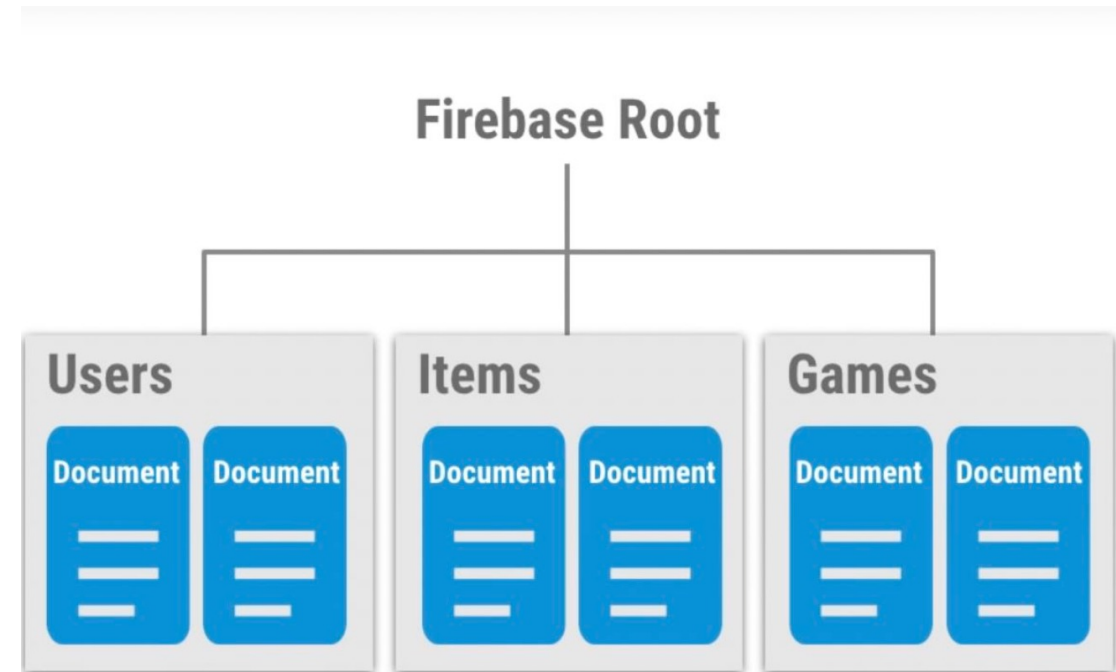
capital: true

Document

```
bird_type: "swallow"
airspeed: 42.733
coconut_capacity: 0.62
isNative: false
icon: <binary data>
vector:
  {x: 36.4255,
   y: 25.1442,
   z: 18.8816}
distances_traveled:
  [42, 39, 12, 42]
```

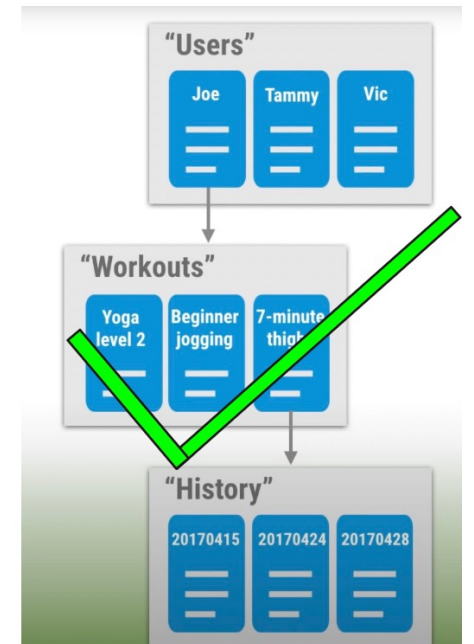
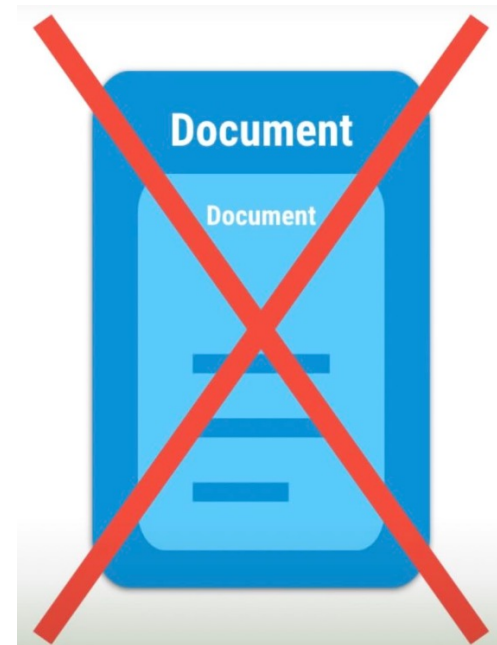
Collections

- A collection contains documents
- Documents in a collection usually represent the same kind of thing
- Firestore does not force every document in a collection to have the same fields
- Keeping a consistent structure makes app code easier to write



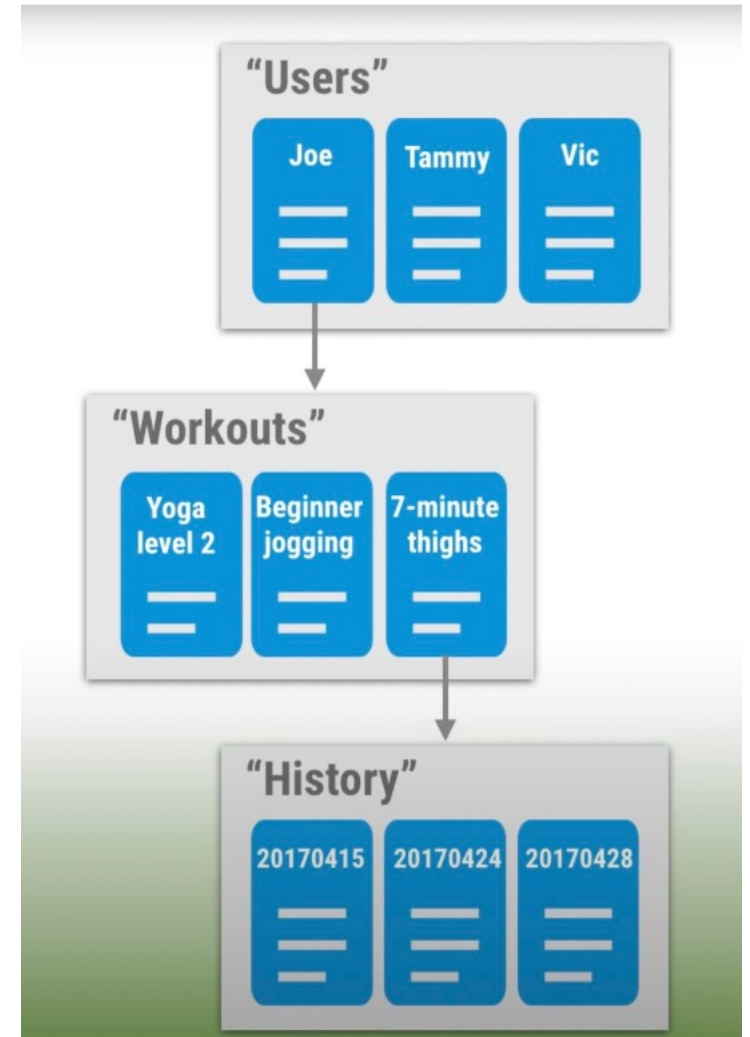
Document Paths

- Firestore data is accessed by path
- Paths alternate between collections and documents
- A document ID only needs to be unique inside its collection
- Example path:
 - Cities/city_001



Nested Data and Subcollections

- Documents can store simple fields, lists, and maps
- Documents cannot directly contain other documents
- A document can have subcollections
- Subcollections are useful when one item has many related child records
- Example:
 - Users/user_123/favourites/city_001



Realtime Updates and Offline Caching

- **Realtime updates**

- app can listen for database changes

- **Offline caching**

- apps can read cached data when offline

- **Queries**

- apps can request selected documents from a collection

- **Pagination**

- apps can load large collections in smaller batches

- Today's Firestore Integration lab uses realtime updates to keep the app UI synchronized with Firestore